

ESIT-2026 Challenge - Task 2

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The task and the fixed metric

Problem

A pretrained CNN f_θ was trained on data with a contaminated subset $\mathcal{F}$ (is_forget=1). **Remove $\mathcal{F}$'s influence without retraining from scratch**, preserving accuracy on the retain set $\mathcal{R}$ and the unseen test set.

Mandatory evaluation — the only freedom is the fine-tune

$$\underbrace{\text{one fine-tuned model}}_{\text{start from baseline}} \rightarrow \underbrace{e_i = \|f_\theta(x_i) - \mathbf{y}_i\|_2}_{\text{vs. original labels}} \xrightarrow{\text{LogReg}} \hat{z} \in \{0, 1\}$$

Single model only — no ensembles / thresholds / cross-fitting in the final path.

Good unlearning $\iff \mathbb{E}_{\mathcal{F}}[e_i] \gg \mathbb{E}_{\mathcal{R}}[e_i] \iff$ the detector separates them.

What is the contamination?

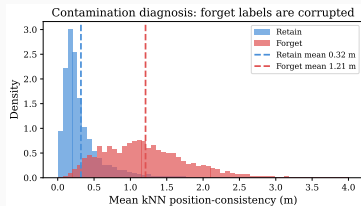
The forget labels are corrupted positions — the CSI is genuine, the stored (x, y) is wrong.

kNN-consistency (label vs. its 10 nearest *retain* neighbours, CSI-PCA space):

	Retain	Forget
mean shift	0.32 m	1.21 m
AUC(z shift)	0.936	

If the model predicts the *true* position,

$e_i \approx \|\mathbf{y}_i^{\text{true}} - \mathbf{y}_i\|_2$ — **error = corruption magnitude**, which transfers to test.



Family A — Relabel: denoise the labels

Idea: give the model a *trusted* target for each forget sample, then fine-tune. The error against the *original* label then equals the corruption.

1. For each forget sample i , find its $k = 10$ nearest **retain** neighbours in the 64-dim PCA of $|\text{CSI}|$.
2. Replace its label with their mean position: $\hat{\mathbf{y}}_i^{\text{corr}} = \frac{1}{k} \sum_{j \in \text{kNN}_{\mathcal{R}}(i)} \mathbf{p}_j$ (retain labels untouched).
3. Fine-tune on the corrected targets.

Why it works for the metric

The model learns the true position $\Rightarrow$ for a forget sample the error vs. the *original* (corrupted) label is large and **transfers exactly** to the test set.

Result: self-MIA 0.927, retain error 0.114 m, forget error 0.861 m.

Family B — Activation-trajectory divergence

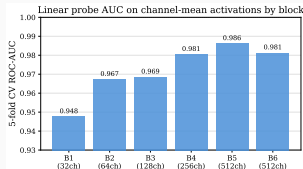
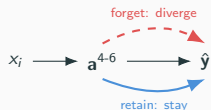
Motivation: the CNN *already* separates forget from retain internally (probe AUC **0.986** at block 5), yet parameter-space dampening (SSD) *fails* (Fisher importance entangled).

So act in activation space.

Method (vs. frozen teacher f_{θ_0}):

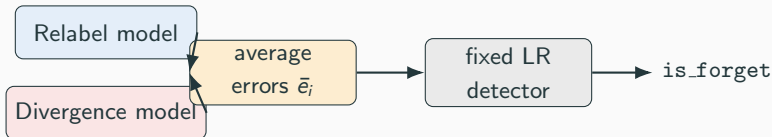
- **Diverge** forget pre-ReLU acts off-trajectory.
- **Anchor** retain post-ReLU acts to the teacher.
- + kNN forget-position loss (transferable signal).

Real representational change, not just a relabel.



The small ensemble

Why ensemble? The two families make *complementary* errors: relabel gives a clean transferable signal; divergence detaches the representation. Averaging their per-sample errors before the detector yields a **more separable** error distribution.



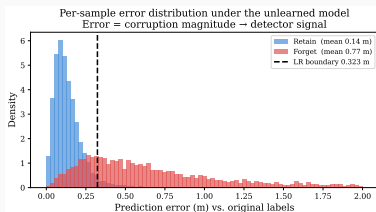
- Members span both families (+ a gradient-ascent variant for diversity).
- Organizers later mandated *single-model* $\Rightarrow$ each family becomes a standalone compliant submission.

Results

Method	self-MIA	retE	fgtE	fgt%
Relabel	0.927	0.114	0.861	0.538
Divergence	0.881	0.139	0.769	0.519
Baseline	—	0.101	0.125	—

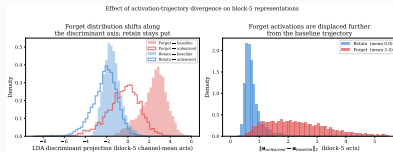
retE / fgtE = mean retain / forget error (m).

fgt% = LR test forget rate (true ≈ 0.50).



Why this is real unlearning — not just relabelling

- Divergence changes *where* the model maps forget inputs in activation space — direct evidence the internal computation changed.
- Forget activations are pushed off the teacher trajectory; retain stays anchored.
- **Ablation:** drop the kNN-position loss and the activation push alone is partly train-specific (worse transfer). The two terms are complementary: *detachment* + *transferable signal*.



Block-5 activations: **forget** displaced after unlearning, **retain** anchored.

Summary

1. **Signal**: contamination = corrupted positions; error = corruption magnitude.
2. **Family A** relabels with kNN-corrected positions (transferable signal).
3. **Family B** diverges forget activations off a frozen teacher (representational).
4. **Small ensemble** of both

Backup: divergence loss & hyperparameters

Total loss (student f_θ , frozen teacher f_{θ_0} , blocks 4–6):

$$\mathcal{L} = L_{\text{pos}} + \beta L_{\text{anc}} + \gamma_t L_{\text{div}} + \nu L_{\text{norm}} + \delta L_{\text{pos},\text{F}}$$

- L_{anc} : retain post-ReLU activations pinned to teacher (relative squared dev.).
- L_{div} : hinge-cosine $\text{ReLU}(\cos(\mathbf{a}_s^\ell, \mathbf{a}_t^\ell))$ on forget pre-ReLU BN.
- L_{norm} : norm-keeper (prevents all-zero collapse). $L_{\text{pos},\text{F}}$: kNN forget position.

β (anchor)	3.0	lr 5×10^{-5} , Adam, batch 64
γ (diverge)	0.5 (ramp 2 ep)	30 epochs, checkpoint ep 24
ν (norm)	0.1	BN running stats frozen
δ (kNN pos)	1.0	select: max self-MIA, retain ≤ 0.25 m

β **calibrates the detector**: tighter anchor $\Rightarrow$ lower train retain error $\Rightarrow$ LR threshold shifts $\Rightarrow$ test forget rate $\rightarrow 0.50$ (cross-checked by an unsupervised GMM split — no leaderboard probing).